

HISTOLOGY OF SMALL INTESTINE

ANATOMY, B.SC MEDICAL TECHNOLOGY

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THE SMALL INTESTINE

Small intestine is about 5 meters long.

It is divided into 3 parts-

- a. DUODENUM (25 cm)
- b. JEJUNUM (2 meters)
- c. ILEUM (3 meters)

It is the principal site for absorption of products of digestion. Digestion is completed in small intestine.

MICROSCOPIC FEATURES

The wall of the intestine has four layers:

- Mucosa
- Submucosa
- Muscular layer
- Serosa

THE MUCOSA MEMBRANE

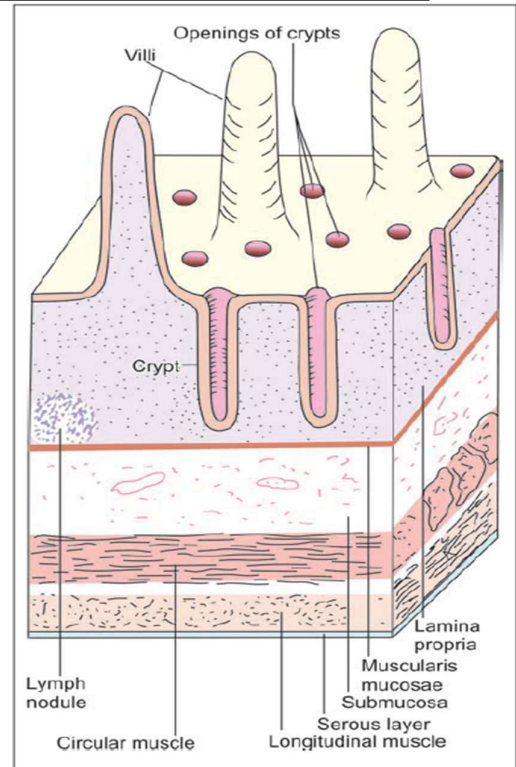
The surface area of mucosa membrane is extensive.

This is achieved by virtue of the following-

- The presence of the numerous circular folds
- The presence of the numerous finger like projections, VILLI that projects from the surface of the mucosa into the lumen.
- The presence of the numerous depressions or CRYPTS that invade the lamina propria.
- The presents of the microvilli on the luminal surface of the cells lining the mucosa.

The mucosa membrane consists of (visceral to parietal)

- a. Lining epithelium
- b. Lamina propria
- c. Muscularis mucosa



CIRCULAR FOLDS

The circular folds are also called VALVES OF KERKRING. Each folds are made up of all the layers of the mucosa (lining epithelium, lamina propria, and muscularis mucosa). The submucosa also extends into the folds. They are absent in the first part of the duodenum and terminal part of the ileum. These folds gradually decrease in number along the length of the small intestine.

Apart from adding extensive surface area to the mucosa of the intestine, it also slows down the passage of contents through it facilitating absorption.

THE VILLI

The villi are typically finger like projections consisting of a core reticular tissue covered by a surface epithelium. The connective tissue core has numerous blood capillaries forming a plexus. The epithelium of the capillaries are fenestrated thus allowing rapid absorption of nutrients into the blood. Each villus contains central lymphatic vessel called **lacteal**. Distally the lacteal binds with tip of the villus, and proximally it ends in a plexus of lymphatic vessels presents in the lamina propria.

THE CRYPTS OF LIEBERKUHN

The crypts are the tubular invaginations of the *epithelium* into the *lamina propria*. They are really simple tubular intestinal glands that are lined by epithelium. The epithelium is supported outside by a basement membrane.

THE EPITHELIUM LINING

The epithelium lining covering villi, mucosa, crypts consists of columnar cells that are specialised in absorption. There are called **enterocytes**. Scattered among the columnar cells, there are mucous secreting **goblet cells**.

The cells lining the crypts are predominantly undifferentiated. These cells multiply to give rise to absorptive columnar cells(**enterocytes**) and to goblet cells. Near the bases of the crypts there are **Paneth cells** that secrete enzymes. Endocrine cells (bearing membrane bound granules filled with various neuroactive peptides) are also present.

ENTEROCYTES

- Brush bordered columnar epithelium
- Has oval nucleus in its lower part
- Microvilli is present and is responsible for the striated appearance
- Microvilli increases the absorptive surface area of the cells
- Each microvilli has a wall of plasma membrane

The cytoplasm of the absorptive cells contains usual organelles, including lysosomes and smooth

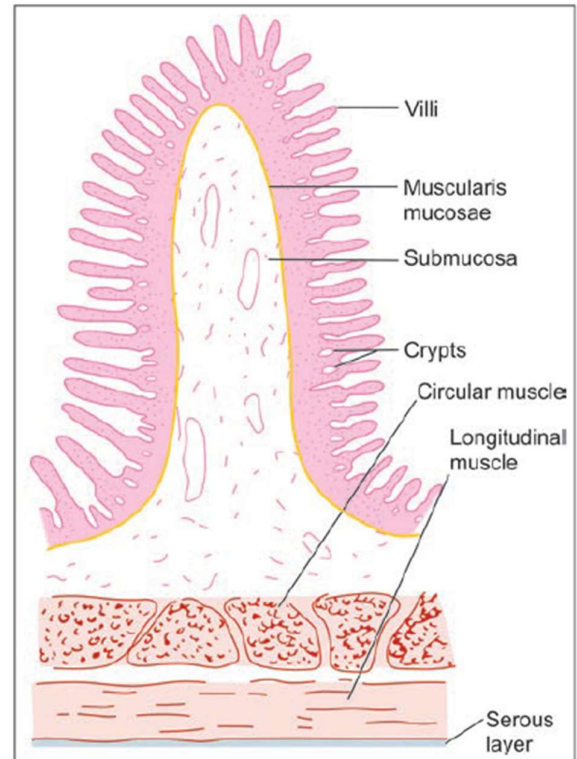


Fig. 16.8: Longitudinal section through a part of the small intestine seen at a very low magnification to show a circular mucosal fold (Schematic representation) Note the submucosa extending into the mucosal fold.

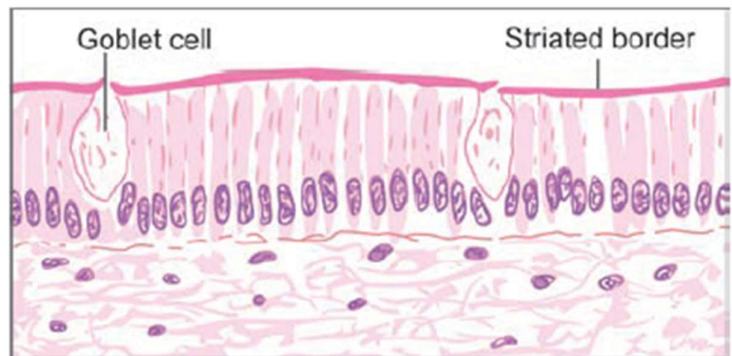


Fig. 16.9: Columnar epithelium lining the small intestine. Note the striated border and some goblet cells (Schematic representation)

ER. These cells are responsible for absorption of amino acids, carbohydrates and lipids present in the digested food.

GOBLET CELLS

Goblet cells has narrow stem attached to the base and expanded upper part that is distended with mucin granules. The nucleus is flattened and present near the base of the cell. Goblet cells are mucous secreting cells. The luminal surface of the goblet cells has irregular microvilli. Mucous cells increase in number as we pass down the small intestine, being few in duodenum and most numerous in terminal ileum.

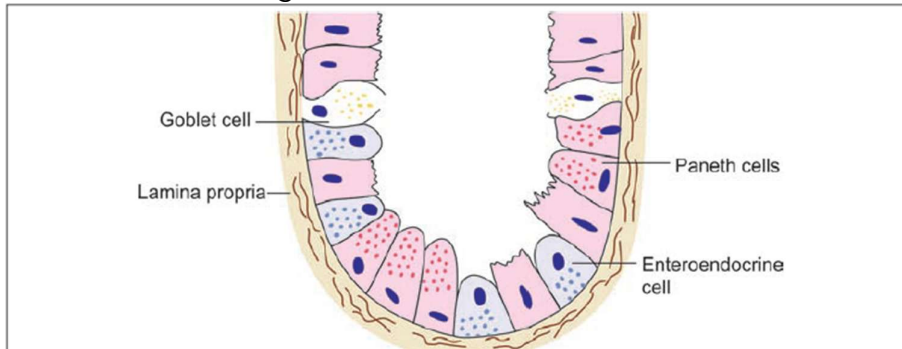


Fig. 16.10: Cells of small intestine (Schematic representation)

UNDIFFERENTIATED CELLS

Undifferentiated cells are found in the crypts of lieberkuhn. These are columnar cells similar to the enterocytes but has underdeveloped microvilli. The cytoplasm contains secretory granules. These cells undergoes mitosis and moves upward to form enterocytes or goblet cells.

PANETH CELLS (ZYMOGEN CELLS)

These cells are found in the deeper part of intestinal crypts. They contain prominent eosinophilic secretory granules. They also has irregular microvilli. Paneth cells are rich in zinc. They are known to produce lysozyme that destroys bacteria.

ENDOCRINE CELLS

Cells containing membrane bound vesicles filled with neuroactive substances are present in the epithelium lining of the small intestine. They are most numerous in the deeper part of crypts.

LYMPHOID TISSUE OF THE SMALL INTESTINE

Solitary and aggregated lymphatic follicles are present throughout the lamina propria of the small intestine. The solitary follicles becomes more numerous and the



Fig. 16.11: High power view of an intestinal crypt showing Paneth cells with red staining supranuclear granules (Schematic representation)

aggregated follicles larger in proceeding along the small intestine. The lymphoid tissue may occasionally extend into the submucosa.

The epithelium overlying the lymphatic follicles contains special follicle-associated epithelial cells or M-cells. These cells are columnar and believed to take up antigens present in the lumen of the intestine and transfer them to the subjacent lymphoid tissue, which can then produce antibodies against the antigens.

PAYER'S PATCHES

Small collections of lymphoid tissue, similar in structure to the follicles of lymph nodes may be present anywhere along the length of the gut. They are called solitary lymphatic follicles. Larger aggregations of lymphoid tissue, each consisting of 10 to 200 follicles are also present in the small intestine. They are called aggregated lymphatic follicles or payer's patches.

- Can be seen with naked eye
- About 200 of them can be counted in the human gut
- May or may not have villi
- 2cm to 10cm in size

DISTINGUISH FEATURES OF DUODENUM, JEJUNUM, ILEUM

DUODENUM

the duodenum is easily distinguished from jejunum and ileum because of the presence of **brunner's gland** (duodenal gland). They are present in the *submucosa*. They are *tubule-alveolar* gland and their duct passes through the *muscularis mucosa* to open into the *intestinal crypts*.

The cells lining the alveoli of duodenal glands are predominantly mucous secreting columnar cells having flattened basal nuclei. Some endocrine cells are also present. They are more numerous in the proximal part of the duodenum.

Secretions- mucous, bicarbonate ion (to neutralise gastric acid entering duodenum) and enzyme that activates trypsinogen produced by the pancreas.

JEJUNUM

As compared to ileum, jejunum has following features:

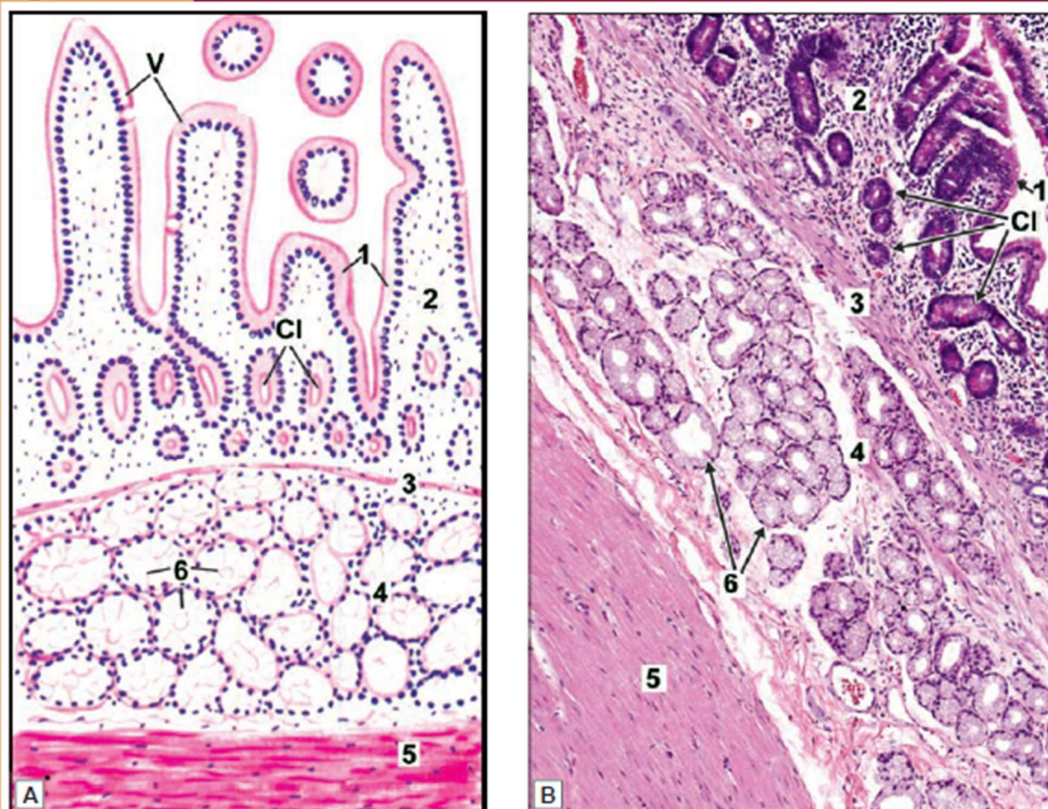
- A larger diameter
- A thicker wall
- Larger and more numerous circular folds
- Larger villi
- Fewer solitary follicles, aggregated follicles are absent in proximal jejunum and small in distal jejunum
- Greater vascularity

ILEUM

The villi are thin and slender in the region of ileum. The submucosa contains payer's patches. M cells are found overlying the lymphoid follicles.



PLATE 16.6: Duodenum



Duodenum. A. As seen in drawing; B. Photomicrograph

Courtesy: Atlas of Histopathology. 1st Edition. Ivan Damjanov. Jaypee Brothers. 2012. p 117

The general structure of the duodenum is the same as that described for the jejunum, except that the submucosa is packed with mucous secreting glands of Brunner.

Note: The intestinal crypts lie 'above' the muscularis mucosae while the glands of Brunner lie 'below' it. The presence of the glands of Brunner is a distinctive feature of the duodenum.

The photomicrograph shows Brunner's glands which are mucous glands present in the submucosa of the duodenum.

Key

- | | |
|---|--------------------------|
| 1. Columnar epithelial lining with goblet cells | 5. Muscularis externa |
| 2. Lamina propria | 6. Glands of Brunner |
| 3. Muscularis mucosa | V. Villi |
| 4. Submucosa with duodenal glands of Brunner | Cl. Crypts of Lieberkuhn |



PLATE 16.5: Jejunum



Jejunum. A. As seen in drawing; B. Photomicrograph

Courtesy: Atlas of Histopathology. 1st Edition. Ivan Damjanov. Jaypee Brothers. 2012. p 117

In this figure we see features of the typical structure of the small intestine. The mucosa shows numerous finger-like projections or villi. Each villus has a covering of columnar epithelium that covers a core of delicate connective tissue. Some goblet cells are also seen. Numerous tubular depressions, or crypts dip into the lamina propria. These crypts are also lined by columnar cells. The mucosa is separated from the submucosa by the muscularis mucosae. The intestine is surrounded by circular and longitudinal layers of smooth muscle which constitute muscularis externa.

Note: A solitary lymph nodule is present in the submucosa in the drawing.

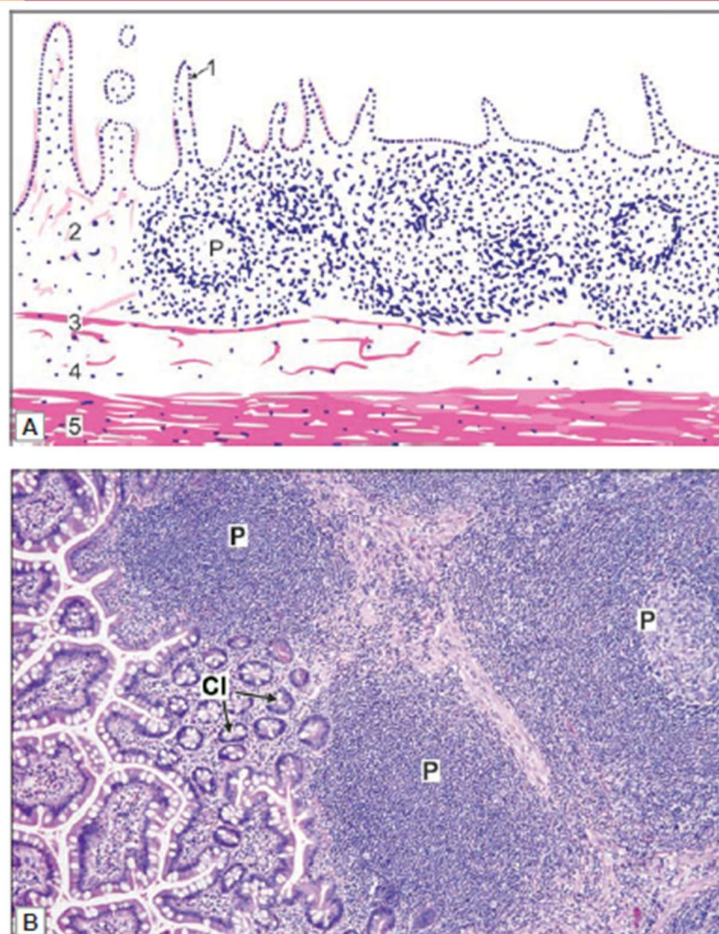
The photomicrograph shows the finger-like villi lined by columnar epithelial cells and goblet cells. The villi dip down into the crypts of Lieberkuhn. At the bases of these are Paneth cells with eosinophilic supranuclear granules.

Key

- | | |
|---|--------------------------|
| 1. Columnar epithelial lining with goblet cells | 6. Lymph nodule |
| 2. Lamina propria | V. Villi |
| 3. Muscularis mucosa | Cl. Crypts of Lieberkuhn |
| 4. Submucosa | |
| 5. Muscularis externa | |



PLATE 16.7: Ileum (Peyer's Patch)



Duodenum. A. As seen in drawing; B. Photomicrograph.

Courtesy: Atlas of Histopathology. 1st Edition. Ivan Damjanov. Jaypee Brothers. 2012. p 117

The general structure of the ileum is similar to that of the jejunum except for

- The entire thickness of the lamina propria is infiltrated with lymphocytes amongst which typical lymphatic follicles can be seen which may extend into the submucosa. These lymphatic follicles are called as Peyer's patches
- In the region overlying the 'Peyer's patch' villi may be rudimentary or absent.

Note: Presence of Peyer's patches is the most distinguishing feature of ileum.

Key

- | | |
|---|--------------------------|
| 1. Columnar epithelial lining with goblet cells | 4. Submucosa |
| 2. Lamina propria | 5. Muscularis externa |
| 3. Muscularis mucosa | P. Peyer's patches |
| | Cl. Crypts of Lieberkuhn |
